

RESEARCH PREPRINT

7/10 Stars Please

Prompt specificity and human preference in the Cadybara Arena pilot

Research question

To what extent did prompt specificity affect preference outcomes for AI-generated CAD designs in the Cadybara Arena pilot?

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Abstract

How much detail should a user place in a text-to-CAD prompt? We examine that question with the Cadybara Arena pilot, which displayed pairs of generated CAD renders and recorded a preference or draw. The archive contains 94 renderable designs from three product families at prompt-detail Levels 1, 3, 5, 7, and 10. We reconstructed 245 archived votes, accepted 230 under the arena's recorded quality rules, and retained 15 flagged votes for sensitivity checks. A level's raw preference score is its wins plus half its draws divided by its exposures.

Level 7 earned the highest raw pooled score: 52.5 preference points across 84 exposures, or 62.5%. Its opponent mix was also the weakest in the archive, with an exposure-weighted opponent score of 46.9%; Level 10 faced the strongest mix at 52.9%. We therefore fitted Bradley–Terry paired-comparison models. Level 7 remained first after schedule adjustment, at 62.9% against an average tested level in a model with product-family effects and 63.9% in a same-family-only fit. The ordering is still imperfect: Level 7 scored only 43.3% in its 15 direct votes against Level 3, while the fitted model expects it to lead. Family subsets were small and disagreed: Level 7 led snowmen, Level 3 led wall hooks, and Level 10 led wall planters. All votes share one stored browser-session identifier, and exact prompts survive only for planters. The archive supports a descriptive Level-7 lead; it does not establish a population preference or a reusable numerical optimum.

Keywords—text-to-CAD, prompt specificity, pairwise preference, crowdsourcing, human evaluation, generative design

1. Introduction

Prompting a text-to-CAD system looks simple until the user has to decide how much to say. A short prompt leaves the model room to choose a form, but it may omit the mounting feature, dimensions, or manufacturing constraints that make the object useful. A highly detailed prompt can communicate those requirements, but it can also force the generator to coordinate more dimensions, operations, and edge conditions at once. In practice, the prompt is both a design brief and an instruction set. Its level of specificity may therefore change not only whether the model follows the request, but also whether a human prefers the resulting design.

This matters more as CAD generation moves from coarse shape synthesis toward editable, parametric models. DeepCAD represented parts as sequences of CAD operations [4]. Text2CAD connected beginner-to-expert descriptions to sequential CAD generation [5], and CAD-Llama adapted language models to structured parametric CAD code [6]. A prompt can now specify a concept, a manufacturing brief, or something close to a construction recipe. Those are different tasks even when they share one label.

The Cadybara Arena pilot was built to compare generated designs through simple pairwise judgments. Two renders appeared side by side, and the interface recorded a left win, a right win, or a draw. This follows the broad logic of human-preference arenas such as Chatbot Arena [1], but the objects here are CAD designs rather than text responses. The archive also records a prompt-specificity level for every renderable design. This creates an opportunity to ask a focused question: **To what extent did prompt specificity affect preference outcomes for AI-generated CAD designs in the Cadybara Arena pilot?**

The archive is small, uneven, and tied to one stored session identity. It also mixes product families in 118 of 230 accepted votes. A simple pooled percentage therefore answers what happened in the arena, but it may reward a level for receiving easier opponents. This is especially important here: Level 7 faced the weakest cross-level schedule and Level 10 the strongest. We use the raw score for transparency, then fit Bradley–Terry models to adjust for the comparison network. We also repeat the model with only the 112 same-family votes so that a preference for snowmen over planters cannot masquerade as a detail effect.

The adjusted result is stronger than a caveat and weaker than a rule. Level 7 remains first in both paired-comparison models. It also loses its direct Level-3 matchup, and the product families point in different directions. Level 1 places second in the pooled archive, which makes the sequence look like one Level-7 spike rather than a smooth dose-response curve. We present the scores as categorical treatment outcomes because each rung adds a different kind of information.

The paper contributes an auditable reanalysis of the pilot. It reconciles every vote, shows schedule strength, reports raw and paired-comparison estimates together, separates mixed- and same-family evidence, and provides the sanitized calculations in a workbook. The result can guide the next experiment, but the present archive cannot tell us what a broader group of users would prefer.

2. Related Work

Pairwise evaluation replaces an absolute score with a relative choice. The Bradley–Terry model estimates a latent strength for each condition from an incomplete network of comparisons [2]. That adjustment matters when some conditions face different opponents. Modern arenas apply related methods at much larger scale. Chatbot Arena combines open pairwise voting with many users and broad prompt coverage [1]. Cadybara borrows the interface logic, although its archived rater and coverage structure is far smaller.

Comparison selection is part of the measurement design. UniCBE treats uniformity and sampling bias as central problems in comparison-based evaluation [3]. The same issue appears in this pilot. A snowman-versus-planter vote contains both a prompt-detail contrast and an object-family contrast. Even within one family, a level can appear more often against weak or strong alternatives. We therefore show the schedule diagnostic, fit the network directly, and report a same-family model.

CAD research also changes what counts as a successful output. DeepCAD generates CAD command sequences [4]. Text2CAD pairs descriptions written at different expertise levels with parametric sequences [5]. CAD-Llama uses a CAD-specific structured representation to generate parametric models [6]. These systems turn prompt wording into part of the experimental treatment rather than incidental interface text.

The newer evaluation benchmarks address different failure modes. CADTests uses executable tests to check whether generated models satisfy geometric and topological prompt requirements [7]. MUSE evaluates code, geometry, and design-intent alignment, with rubrics for functionality, manufacturability, and assemblability [8]. Text2CAD-Bench contains 600 human-curated examples across four complexity levels and pairs non-expert geometric prompts with expert-style procedural prompts [9]. The distinctions are relevant to this study: two prompts can differ in kind as well as length, and a visually preferred render can still fail an engineering requirement.

Our figures follow established advice on direct labeling, common scales, and keeping visual emphasis proportional to the evidence [11], [12]. We avoid fitted curves across Levels 1, 3, 5, 7, and 10 because the numbers name prompt bundles rather than equal units of a continuous dose.

3. Dataset and Arena

3.1 Renderable design set

The public dataset contains 94 active, renderable designs: 33 snowmen, 31 wall hooks, and 30 wall planters. Every item stores a product family, seed label, prompt-specificity level, repetition, experiment label, generation latency, and validation metadata. The tested specificity conditions are Levels 1, 3, 5, 7, and 10. Item counts are not perfectly balanced: Level 1 has 20 designs, Level 3 has 20, Level 5 has 19, Level 7 has 18, and Level 10 has 17. Within the wall-planter family, each tested level has six designs; the snowman and hook counts vary slightly.

All 94 public items are marked renderable and valid. The set excludes failed generation attempts and therefore represents survivors of the generation pipeline. Among those survivors, Level 10 required 2.24 attempts per item on average; 7 of 17 items needed at least three attempts. Level 7 averaged 1.83 attempts, with 2 of 18 needing at least three. The extra retries make Level 10's low pooled preference more striking, since only its successful outputs reached the arena. They still do not reveal the condition's full failure rate.

The exact wall-planter ladder is preserved and reproduced in Appendix A. Figure 1 shows one fixed wall-planter generation run at the five tested levels. The images make the treatment concrete: Level 1 asks only for a planter that

can go on a wall; intermediate levels add mounting, shape, dimensions, and wall thickness; Level 10 specifies detailed geometry, drainage, a back panel, a keyhole slot, fillets, watertightness, and print orientation. The same exact prompt wording is not available for the archived snowman and wall-hook ladders. Those families are analyzed using their preserved level metadata, and we do not reconstruct missing text from titles or memory.

The ladder also explains why the level number should be handled carefully. Level 3 adds a mounting concept, Level 5 fixes shape and orientation, and Level 7 introduces numeric dimensions and printable wall thickness. Level 10 changes the character of the task again by prescribing topology and exact placements. These are cumulative design decisions, not equally spaced increases in word count. The tested labels are retained because they were used during generation, while the analysis treats them as five named conditions.

A matched wall-planter run across the five tested detail levels

Same experiment and repetition; only the prompt-specificity condition changes.



Figure 1. Matched wall-planter examples at the five tested prompt-detail levels from one fixed generation run. The treatment changes from a minimal wall-planter request to a fully specified printable geometry; image selection is matched by experiment and repetition.

3.2 Vote archive and collection

The archive contains 245 votes collected through a public crowdsourcing interface. Each record identifies the left and right item, the outcome, timestamps, and quality-control fields. The recorded rules accept 230 votes and flag 15. We use the accepted set for the main analysis and return to all 245 in the sensitivity appendix. No archived vote is marked as a test vote.

“Crowdsourcing” describes the interface and collection route. It does not describe the evidence stored in this archive. Every vote shares one browser-session identifier, and 222 of the 245 votes were recorded on July 2. The files do not identify the person behind that session, show whether the author participated, or record whether the rater knew the level labels. We therefore treat the archive as one session of repeated choices. The 230 accepted votes are comparison events, not independent respondents.

The accepted votes span 230 unique item pairs, with no repeated pair. Of these, 112 compare designs from the same family and 118 compare different families. For the treatment question, the useful coverage frame is family by cross-level pair. Five tested levels create ten cross-level pairs per family, or 30 cells. All 30 cells appear in the archive, but each contains only one to seven votes. The 97 same-family, cross-level votes average 3.2 votes per cell. This is enough to connect the level network and fit a paired-comparison model; it is thin evidence for family-specific claims.

Collection occurred in three calendar-day groups: 12 raw votes on June 22, 11 on June 23, and 222 on July 2. The pair-selection logic changed before the large July 2 phase. We compare the full accepted archive with the 208 accepted July-2 votes to check whether combining the two collection phases creates the observed ordering.

Table 1. Archive and renderable-design summary

Measure	Value	Interpretation
Renderable designs	94	33 snowmen; 31 hooks; 30 planters
Items by tested level	20 / 20 / 19 / 18 / 17	Levels 1 / 3 / 5 / 7 / 10
Archived votes	245	All preserved for sensitivity
Primary accepted votes	230	Used for main results

Measure	Value	Interpretation
Flagged votes	15	Excluded from primary; retained in raw check
Same-family / mixed-family	112 / 118	Accepted votes
Unique accepted item pairs	230	No accepted pair repeats
Family × cross-level cells	30 / 30	97 votes; 1–7 votes per cell
Archived session identifiers	1	Rater identity and independence unverified
Accepted July 2 votes	208	Collection-phase sensitivity sample

4. Methods

4.1 Raw preference score

An exposure is one appearance of a level in one vote. A win contributes 1 preference point to that exposure, a draw contributes 0.5, and a loss contributes 0. The raw score is

Preference score = (wins + 0.5 × draws) / exposures.

Each accepted vote creates two exposures, so the 230-vote sample contains 460. This formula handles draws transparently and reconciles across the archive: total wins equal total losses, and every draw contributes one point split across two designs. We report both the numerator and denominator when a subset is small.

The live Elo leaderboard is excluded from the evidence. Its initial ratings use validation status, prompt level, title terms, attempt count, latency, and deterministic jitter. Those priors serve the arena's item-selection system, but a level-adjusted starting rating cannot independently establish a level effect.

4.2 Schedule-adjusted paired-comparison models

The raw score depends on which opponents a level faced. We first calculate a descriptive schedule diagnostic: for every cross-level exposure, we assign the opponent's raw pooled score and average those values within the focal level. This diagnostic is not an independent quality estimate because the opponent scores come from the same archive. It simply shows whether the comparison mix was easier or harder on the observed scale.

We then fit Bradley–Terry logistic models [2]. If a vote compares a left and right design, the model represents the log-odds of a left preference as the difference between their latent strengths. A left win is coded 1, a right win 0, and a draw 0.5. The draw treatment matches the raw preference-point definition. Abilities are centered to have mean zero, and we report the logistic transform of each level's centered ability: the modeled probability of beating an artificial average-ability tested level.

This scale keeps the adjusted estimates comparable to the raw percentages while changing their meaning. A reported 62.9% does not mean that Level 7 won 62.9% of a new sample. It means that the fitted level coefficient corresponds to a 62.9% chance against a hypothetical opponent whose level ability is the average of the five tested coefficients, holding family constant. Pairwise fitted probabilities come from the difference between two level coefficients. The model uses the entire connected comparison network, so a condition can receive a high fitted strength even when one direct matchup runs against the global ordering.

Two specifications address the main confounds. The first uses all accepted votes and estimates additive effects for prompt level and product family. A snowman-versus-planter vote can inform the family coefficients while still contributing to the connected level network. The second uses only same-family votes, so family effects cancel by construction. Of 112 same-family votes, 97 compare different levels and identify the level strengths. We report point estimates only. With one stored session identity, standard errors or bootstrap bounds would describe resampling of one person's comparison events rather than uncertainty across raters. A vote-level bootstrap can be computed mechanically [10], but we do not use it as inferential support.

The Bradley–Terry model assumes one transitive strength per level. We check that assumption against observed Level-7 head-to-head outcomes. A disagreement between a fitted probability and a direct pair is reported as model lack of fit, not smoothed away.

4.3 Family-specific and categorical comparisons

We calculate raw scores separately for snowmen, wall hooks, and wall planters using same-family votes. These subsets preserve the recorded comparison outcomes but remain small: 36 snowman votes, 36 hook votes, and 40 planter votes. We state wins, draws, losses, or preference points alongside percentages when the denominator is especially small.

Levels 1, 3, 5, 7, and 10 are categorical prompt bundles. Their numerical spacing is uneven, and each step adds a different type of information. We therefore compare named conditions and do not fit a continuous trend, polynomial, or dose-response curve. The preserved planter ladder permits a semantic interpretation of those bundles. The snowman and hook labels cannot be audited against their missing prompt text.

4.4 Sensitivity, coverage, and reproducibility

Three archive views test the recorded filtering choices: all 245 raw votes, all 230 accepted votes, and the 208 accepted July-2 votes. For each view we report raw scores and refit the level-plus-family Bradley–Terry model. This checks whether the ordering depends on the 15 flags or the earlier collection phase.

Coverage is tabulated across the 30 family-by-cross-level cells. A cell counts accepted same-family votes between one pair of tested levels. Within-level votes are retained in the workbook but do not identify a difference between level strengths. The figure makes the sparse cells visible instead of comparing the archive with thousands of item pairs that the treatment analysis never intended to cover.

All calculations were rebuilt from archived item and vote files. The workbook contains sanitized item keys, every archived vote, a two-row-per-vote exposure table, formula-derived raw summaries, paired-comparison coefficients and probabilities, sensitivity results, coverage tables, and figure source values. It excludes browser-session identifiers, vote and battle IDs, network fields, private hashes, and source hashes.

5. Results

5.1 Level 7 remained first after schedule adjustment

Table 2 and Figure 2 show the main comparison. Level 7 earned 52.5 preference points across 84 exposures: 47 wins, 11 draws, and 26 losses, for a raw score of 62.5%. Level 1 was second with 54.5 points across 99 exposures (55.1%). Level 3 scored 50.0%, Level 5 42.7%, and Level 10 40.1%.

The raw sequence is 55.1%, 50.0%, 42.7%, 62.5%, and 40.1%. It has one spike rather than a smooth peak. Level 1's second-place result also matters: the minimal condition beat Level 5 and Level 10 directly and remained competitive overall. More detail did not reliably improve preference in this archive.

The schedule was uneven in the same direction as the headline. Level 7's cross-level opponents averaged 46.9% on the raw pooled scale, the weakest mix of the five conditions. Level 10's opponents averaged 52.9%, the strongest. The difference is large enough that an unadjusted claim would be incomplete.

Table 2. Raw and schedule-adjusted results by prompt-detail level

Level	Points / exposures	Raw score	Mean opponent raw	BT all + family	BT same-family
Level 1	54.5 / 99	55.1%	48.4%	53.8%	53.7%
Level 3	47.5 / 95	50.0%	49.6%	49.7%	48.9%
Level 5	41.0 / 96	42.7%	51.4%	41.5%	41.0%
Level 7	52.5 / 84	62.5%	46.9%	62.9%	63.9%
Level 10	34.5 / 86	40.1%	52.9%	41.8%	42.4%

Level 7 remained first after schedule adjustment

Point estimates only. Prompt levels are categorical conditions; markers are not connected as a trend.

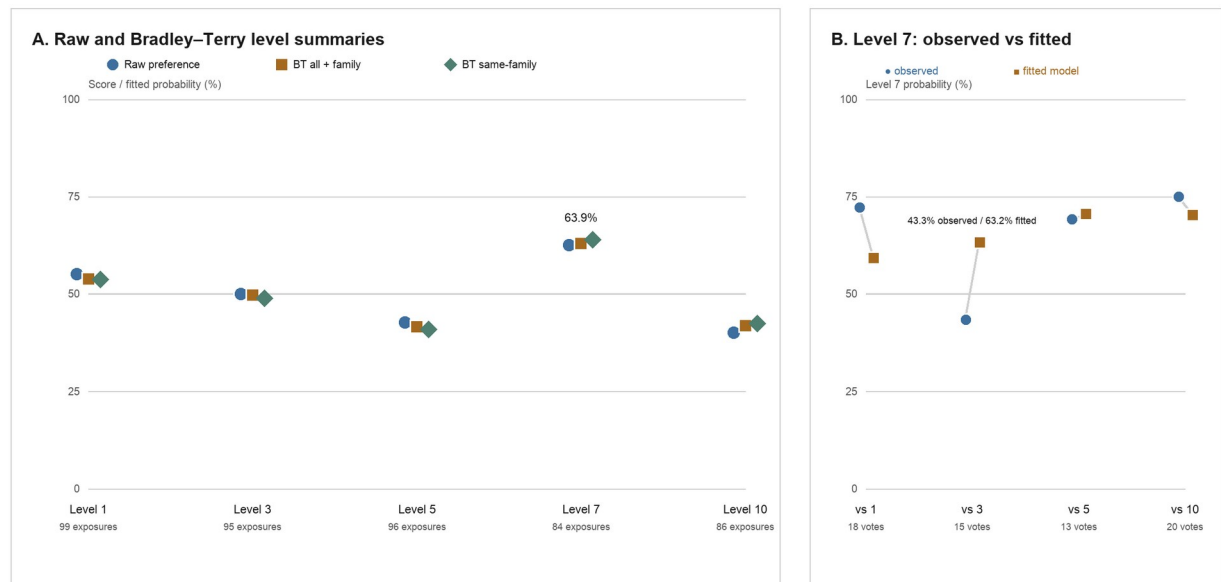


Figure 2. Level 7 remains first in the raw, family-adjusted, and same-family-only summaries. The direct Level-3 comparison runs in the opposite direction: 43.3% observed for Level 7 versus 63.2% fitted. Markers are point estimates; panel A lists exposures and panel B lists votes.

Both paired-comparison fits preserve the Level-7 lead. In the all-vote model with product-family effects, Level 7 has a modeled 62.9% probability against the average tested level. The same-family-only fit gives 63.9%. Level 1 is again second at 53.8% and 53.7%, respectively. Level 10 rises slightly after adjustment, from 40.1% raw to 41.8% in the family-adjusted model and 42.4% in the same-family model, but it remains fourth or fifth depending on the specification. The schedule correction therefore narrows the raw contrast without reversing the ranking.

5.2 The Level-3 reversal exposes model tension

Level 7 performed strongly against three alternatives. It earned 13.0 points in 18 direct votes against Level 1 (72.2%), 9.0 in 13 against Level 5 (69.2%), and 15.0 in 20 against Level 10 (75.0%).

Against Level 3, it earned 6.5 points in 15 votes: 6 wins, 1 draw, and 8 losses, or 43.3%. The family-adjusted Bradley-Terry model predicts 63.2% for that matchup because Level 7 performs better elsewhere in the network. The 19.9-point observed-versus-fitted gap is a clear warning about the model's transitivity assumption. The adjusted ranking summarizes the whole network; it does not erase the direct loss.

Table 3. Level 7 in direct comparisons

Pair	Votes	W-D-L	Observed	BT fitted	Observed – fitted
Level 7 vs 1	18	11-4-3	72.2%	59.3%	+12.9 pp
Level 7 vs 3	15	6-1-8	43.3%	63.2%	-19.9 pp
Level 7 vs 5	13	9-0-4	69.2%	70.5%	-1.3 pp
Level 7 vs 10	20	13-4-3	75.0%	70.3%	+4.7 pp

The Level-3 row runs against the fitted transitive ordering and is reported as model lack of fit.

5.3 Product family changed the preferred level

Figure 3 shows the same-family raw scores without connecting them as a trend. Snowman Level 7 earned 10.0 points across 15 exposures (66.7%), narrowly above Level 1's 11.5 across 18 (63.9%). Snowman Level 10 earned only 2.5 across 15 (16.7%).

Wall-hook Level 3 earned 12.0 points across 18 exposures (66.7%). Level 7 earned 7.5 across 12 (62.5%). Four percentage points separate the leaders, and the underlying comparison cells contain only one to five votes. The ordering should be read as an exploratory lead.

Wall-planter Level 10 earned 10.0 points across 12 exposures: 9 wins, 2 draws, and 1 loss, or 83.3%. Level 7 earned 9.5 across 15 (63.3%). Two reversed votes would materially change the gap, so the planter result is suggestive rather than stable. It is still the only family with auditable prompt text, and its direction contradicts a universal Level-7 optimum.

Small same-family subsets selected different leaders

Filled circles show raw preference; open squares show family-specific Bradley–Terry estimates. Point estimates only.

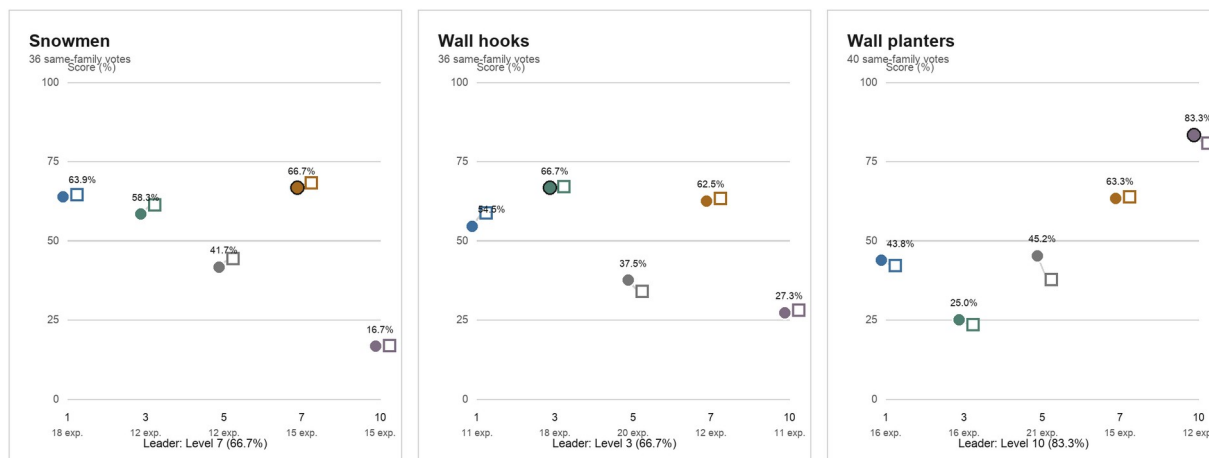


Figure 3. Small same-family subsets produced different leaders: Level 7 for snowmen, Level 3 for wall hooks, and Level 10 for wall planters. Filled circles show raw preference; open squares show family-specific Bradley–Terry point estimates on a common 0–100% scale.

The family-specific Bradley–Terry fits reproduce the same leaders: Level 7 for snowmen, Level 3 for hooks, and Level 10 for planters. Their role is descriptive because each family contributes only 29 to 37 informative cross-level votes. The missing hook and snowman prompt ladders create a deeper problem: those two family analyses compare metadata labels whose semantic treatment cannot be independently checked.

5.4 The comparison network is connected and thin

Figure 4 displays the 30 same-family cross-level cells. Every cell is represented, which connects all five levels inside each product family. Counts range from one to seven votes. Eleven cells have only one or two votes, and the average is 3.2. Coverage exists in the combinatorial sense, but the individual family-by-pair estimates are fragile.

All family-by-level cells were touched, usually only a few times

Counts show accepted same-family votes. Each column is one cross-level treatment comparison.

	1 vs 3	1 vs 5	1 vs 7	1 vs 10	3 vs 5	3 vs 7	3 vs 10	5 vs 7	5 vs 10	7 vs 10
Snowmen	3	5	4	4	1	3	3	2	2	4
Wall hooks	5	2	2	2	5	1	3	3	4	2
Wall planters	2	7	5	2	5	3	4	3	2	4

97 votes across 30 cells · mean 3.2 · range 1–7 · 11 cells contain one or two votes

Figure 4. Same-family coverage across the 30 family-by-cross-level cells. Every cell is represented, but counts range from one to seven votes and 11 cells contain only one or two.

No item pair repeats, so the archive cannot measure agreement on an identical choice. A stronger design would allocate the same number of repeated comparisons to each family-by-level cell and distribute those repeats across separately identifiable, blinded raters. The present heatmap is useful because it shows exactly where those repeats are needed.

5.5 Filtering and collection phase do not change the first-ranked level

The raw full archive produces scores of 54.7%, 49.5%, 43.3%, 61.6%, and 39.7% for Levels 1, 3, 5, 7, and 10 respectively. Filtering to the 230 accepted votes raises Level 7 to 62.5% and gives the primary values reported above. Restricting further to the 208 accepted July 2 votes produces 55.7%, 50.6%, 42.5%, 61.5%, and 40.9%. In every view, Level 7 remains the pooled leader and Level 10 remains below 41%.

The family-adjusted Bradley–Terry estimate for Level 7 is 62.4% in the raw archive, 62.9% in the accepted archive, and 61.8% in the accepted July-2 sample. These checks show that the first-ranked level is not created by the 15 flags or the 23 early votes. All three samples still come from the same stored session identity.

6. Discussion

6.1 Interpreting the Level-7 lead

The Level-7 result is consistent with a balance between missing information and excessive constraint. In the preserved planter ladder, Level 7 specifies the cylindrical form, outer dimensions, wall thickness, open top, flat back, and mounting hole. Levels 8 through 10 add drainage layout, exact panel geometry, slot construction, fillets, watertightness, and print orientation. Level 7 therefore marks a recognizable design brief: the main functional geometry is present while several implementation details remain open.

That story is plausible, but the archive never measures prompt adherence, violated constraints, cognitive load, or geometric error. The planter subset also points the other way, with Level 10 leading. Additional detail may help when it supplies a necessary interface and hurt when it creates brittle coordination demands. The present data cannot separate those mechanisms.

Level 1 deserves equal attention. A minimal prompt placed second overall and remained second after both Bradley–Terry adjustments. The result suggests that the generator sometimes benefits from freedom, or that the rater valued clean form over instruction coverage. Since prompt adherence was never scored, those explanations remain open.

6.2 Prompt levels need semantic definitions

The five labels are categories, not equal steps on a ruler. The planter ladder adds mounting intent, size, shape, dimensions, material context, drainage, topology, and exact feature placement at different rungs. A shift from Level 5 to Level 7 does not represent the same treatment increment as a shift from Level 7 to Level 10.

Future ladders should record the information type added at each step: identity, function, interface, dimensions, process, topology, and feature placement. A balanced experiment can then test both amount and kind of specificity. The supplied library of rings, plates, brackets, pipe fittings, enclosures, complex parts, and modifiers is suitable material for that follow-up. It was not used to generate the present archive.

The missing snowman and hook prompts prevent semantic comparison across families. Their metadata can reproduce the arena's internal labels, but it cannot show that “Level 7” meant the same amount or kind of detail. Recovering or regenerating those ladders is required before the family interaction can become a treatment claim.

6.3 Preference should be paired with technical evaluation

Render preference captures coherence, proportion, apparent usability, and finish. It cannot verify topology, dimensions, tolerances, drainage, fastener fit, or printability. CADTests addresses geometric and topological prompt

requirements with executable tests [7]. MUSE adds code, geometry, and design-intent checks with engineering rubrics [8]. Text2CAD-Bench shows that model performance falls as topology and feature complexity increase across its four-level benchmark [9].

A stronger evaluation would combine executable prompt checks, manufacturability review, and blinded human preference. These measures can disagree in useful ways. A Level-10 planter might satisfy more requirements while a Level-7 render looks cleaner. Recording both outcomes would turn the under-specification-versus-overconstraint interpretation into a testable mechanism.

6.4 A replication that can answer the question

The next arena should schedule comparisons rather than accept whatever pair mix happens to arrive. For each family, all ten cross-level pairs should receive the same target number of votes. A smaller set of item pairs should repeat across raters so agreement and item-specific noise can be estimated. Left-right order should be randomized, level labels hidden, and every rater should receive a balanced subset rather than hundreds of consecutive choices.

Five raters would already improve the design if each completed a manageable block and the blocks overlapped. Separate anonymous rater codes could preserve privacy while supporting per-rater summaries, agreement measures, and leave-one-rater-out checks. The study should record whether the author participated and document any training examples shown before voting.

Generation evidence needs the same discipline. Every attempt should be retained with its prompt, model identity, seed, failure status, and validation output. The recovered snowman and hook ladders should follow the same semantic schema as the planter ladder. With those records, the analysis could distinguish generation success, prompt adherence, technical quality, and human preference instead of conditioning everything on a render that survived to the arena.

7. Limitations

All votes share one browser-session identifier. The archive does not establish the number or identity of raters, the author's participation, blinding to prompt level, or independence between choices. The results describe one stored session. They cannot support a population preference, between-rater agreement, or conventional inferential uncertainty.

Treatment integrity is complete only for wall planters. Snowmen and wall hooks retain numerical level metadata without the exact prompts that created those labels. Two-thirds of the family analysis therefore relies on an unaudited independent variable. The planter ladder can be interpreted semantically; the other ladders must be recovered or regenerated.

The comparison network is connected but lightly sampled. All 30 family-by-cross-level cells appear, yet 11 contain only one or two votes. No item pair repeats. Bradley–Terry adjusts for opponent mix by assuming a transitive level strength, and the Level-7-versus-Level-3 reversal shows that assumption is imperfect. The fitted ranking is a compact summary of the archive, not a substitute for balanced random assignment.

Only renderable designs enter the public dataset. Failed attempts are missing, while successful Level-10 items show more retries than Level-7 items. Preference is therefore conditional on surviving generation. The archive cannot calculate attrition or determine whether retries changed the distribution of visible quality.

Raters saw rendered views rather than CAD files or fabricated objects. They could not verify dimensions, topology, tolerances, drainage, screw fit, support requirements, or material behavior. The outcome is visual preference under the arena presentation.

The data use one generic model label. Prompt specificity may interact with model architecture, scale, decoding, or CAD representation, none of which can be estimated here.

Levels 1, 3, 5, 7, and 10 are unevenly spaced bundles, and the missing conditions never entered the arena. The analysis can identify the highest observed category among those five. It cannot locate a numerical optimum along a continuous specificity scale.

8. Conclusion

Level 7 received the highest preference score in the accepted Cadybara archive: 52.5 points across 84 exposures, or 62.5%. It also faced the weakest raw opponent mix. After correcting the comparison network, Level 7 remained first at 62.9% against the average tested level in a family-adjusted Bradley–Terry model and 63.9% in a same-family-only fit. The title “7/10 Stars Please” refers to that repeated descriptive lead.

The archive stops short of a general prompting rule. Level 7 lost its direct Level-3 matchup, Level 1 placed second, and the three families selected three different leaders. Exact prompts survive only for planters, where Level 10 led. One stored session identity prevents population or agreement claims.

The next study should use complete ladders, randomize and repeat every same-family level pair, recruit several separately identifiable and blinded raters with appropriate privacy safeguards, retain failed generation attempts, and score prompt adherence and manufacturability alongside preference. Those changes would test whether moderate detail genuinely balances freedom and constraint. The current paper establishes the pattern worth testing: Level 7 led this archive, even after schedule adjustment.

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Appendix A. Complete wall-planter prompt ladder

The full ladder is reproduced exactly from the preserved study file. Only Levels 1, 3, 5, 7, and 10 were represented in the arena dataset analyzed in this paper. Levels 2, 4, 6, 8, and 9 are included for completeness but do not contribute evidence.

Level 1 — Minimal (tested)

Make me a planter I can put on my wall.

Level 2 — Basic Form Intent (not tested in arena)

Design a wall-mounted planter that holds a small potted plant.

Level 3 — Mounting Concept (tested)

Design a wall-mounted planter that holds a small potted plant and can be hung on a nail or screw in drywall.

Level 4 — Rough Size (not tested in arena)

Design a wall-mounted planter about the size of a coffee mug, that holds a small potted plant and can be hung on a nail or screw in drywall.

Level 5 — Shape And Orientation (tested)

Design a cylindrical wall-mounted planter about the size of a coffee mug, open at the top, that holds a small potted plant and hangs flat against the wall on a single screw.

Level 6 — Approximate Dimensions (not tested in arena)

Design a cylindrical wall-mounted planter roughly 90 mm tall and 80 mm in diameter, open at the top, with a flat back that hangs against the wall using a single screw through a mounting hole.

Level 7 — Wall Thickness And Material Context (tested)

Design a cylindrical 3D-printable wall-mounted planter, 90 mm tall and 80 mm outer diameter, with a 3 mm wall thickness, an open top, and a flat back panel containing one mounting hole sized for a #8 screw.

Level 8 — Drainage And Functional Features (not tested in arena)

Design a cylindrical 3D-printable wall-mounted planter, 90 mm tall and 80 mm outer diameter, with a 3 mm wall thickness, an open top, a flat back panel, one keyhole-shaped mounting slot for a #8 screw centered on the back panel, and four 3 mm drainage holes in the bottom.

Level 9 — Geometric Placement Of Features (not tested in arena)

Design a cylindrical 3D-printable wall-mounted planter. Outer diameter 80 mm, height 90 mm, wall thickness 3 mm. The back is a flat panel tangent to the cylinder, 100 mm tall x 90 mm wide, extending 5 mm above the cylinder top. Centered on the upper portion of the back panel, 15 mm from its top edge, is a keyhole-shaped mounting slot (8 mm circle on top, 4 mm wide x 10 mm tall slot below) sized for a #8 screw. The bottom of the cylinder has four 3 mm drainage holes arranged in a square pattern, each 15 mm from the central axis.

Level 10 — Fully Specified (tested)

Design a 3D-printable wall-mounted planter with the following specifications. The body is a hollow cylinder, 80 mm outer diameter, 90 mm tall, with a uniform 3 mm wall thickness, open at the top and closed at the bottom. The bottom of the cylinder contains four 3 mm diameter drainage holes arranged in a square pattern, each centered 15 mm from the cylinder's central axis. The back is a flat rectangular panel, 100 mm tall x 90 mm wide x 4 mm thick, tangent to the outside of the cylinder along a vertical line, and extending 5 mm above the cylinder's top edge and 5 mm below its bottom edge. The back panel contains one keyhole-shaped mounting slot centered horizontally, with its top edge 15 mm below the top of the back

panel: an 8 mm diameter circular opening with a 4 mm wide x 10 mm tall rectangular slot extending downward from the circle's center, sized to accept a #8 pan-head screw. All external edges except the open top rim have a 1 mm fillet. The model is a single watertight solid suitable for FDM 3D printing without supports when oriented with the back panel flat on the print bed.

Appendix B. Sensitivity results

Table 4. Archive-processing sensitivity results

Sample	Votes	L1 raw	L3 raw	L5 raw	L7 raw	L10 raw	L7 BT
Raw full archive	245	54.7%	49.5%	43.3%	61.6%	39.7%	62.4%
Accepted full archive	230	55.1%	50.0%	42.7%	62.5%	40.1%	62.9%
Accepted July 2 only	208	55.7%	50.6%	42.5%	61.5%	40.9%	61.8%

Filtering and collection phase did not change the first-ranked level

Grouped points show categorical conditions. Left: raw preference. Right: family-adjusted Bradley–Terry estimates.



Figure 5. Raw and family-adjusted sensitivity results for the full archive, accepted archive, and accepted July-2 sample. Level 7 remains first under every archive-processing view.

The raw sample retains all 245 archived votes. The accepted full archive removes 15 flagged votes, and the July-2 sample also removes the earlier collection phase. Level 7 remains first in the raw scores and in the family-adjusted Bradley–Terry estimates for all three samples. This appendix tests archive-processing choices only; every sample retains the same session-identity limitation.